B.Tech CSE - Third Year | Lab 8

Assignment 8: Classes and Objects • *Aim: Define classes, fields and methods, then create objects.*

Experiment No. 8.1: Create and display one Student

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Student {
    int roll;
    String name;
    void show() {
        System.out.println("Roll: " + roll);
        System.out.println("Name: " + name);
    }
}
public class Task_01_Create_and_display_one_Student {
    public static void main(String[] args) {
        Student s = new Student();
        s.roll = 5;
        s.name = "Rohan";
        s.show();
    }
}
```

3. **Output:**

```
Roll: 5
Name: Anand
```

4. **Viva-Voce:** What is a class? What is an object? What is a method?

Experiment No. 8.2: Create and display two Students

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Student {
    int roll;
    String name;
    void show() {
        System.out.println("Roll: " + roll);
        System.out.println("Name: " + name);
    }
}
public class Task_02_Create_and_display_two_Students {
    public static void main(String[] args) {
        Student s1 = new Student();
        s1.roll = 5; s1.name = "Rohan";
        Student s2 = new Student();
        s2.roll = 8; s2.name = "Meera";
        s1.show();
        s2.show();
    }
}
```

3. **Output:**

```
Roll: 5
Name: Rohan
Roll: 8
Name: Meera
```

4. **Viva-Voce:** Can two objects have different field values? What does new do? How is a method called on an object?

Experiment No. 8.3: Check whether a Book is long

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
class Book {
    int pages;
    boolean isLong() {
        return pages > 300;
    }
}
public class Task_03_Check_whether_a_Book_is_long {
    public static void main(String[] args) {
        Book b1 = new Book(); b1.pages = 450;
        Book b2 = new Book(); b2.pages = 120;
        System.out.println(b1.isLong());
        System.out.println(b2.isLong());
    }
}
```

3. **Output:**

```
true
false
```

4. **Viva-Voce:** What is a boolean? What does return do? Why is 450 considered long?